

Real-Time Soccer Monitoring with Apache Spark Structured Streaming

Workshop Presentation

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Project Overview

Real-time soccer monitoring with Apache Spark Structured Streaming.

Dataset: DEBS 2013 Soccer Monitoring dataset.

Static timestamped data is reconstructed as a simulated live stream, in event-time order.

Final goal: simulate a live soccer data stream, process it incrementally and run analysis on it

Motivation and Dataset

Motivation

Soccer tracking data is naturally temporal.

We can interpret sensor records as continuous data stream.

Processing soccer games live is a real-life necessity.

Raw schema

Column	Meaning
sid	Sensor ID
ts	Timestamp
x,y,z	Position
v_abs	Velocity
a_abs	Acceleration
vx,vy,vz	Velocity direction
ax,ay,az	Acceleration direction

System Architecture

Raw DEBS Data → Cleaning → Chunking → Replay → Spark Streaming → Live Output

Step	Script / Folder
Cleaning	src/clean_data.py
Chunking	src/split_data.py
Replay	src/replay_full_game.py
Input stream	data/stream_input/
Output	visualization data/output/

Setup

`config.py`: stores timestamps, paths, chunk size, and schemas.

`clean_data.py`: reads raw data, filters match intervals and adds additional columns.

`split_data.py`: partitions cleaned data by `matchSecond` into chunks.

`replay_full_game.py`: copies chunks into `data/stream_input/..`

Demo

Next Steps

Add live indicators such as ball possession,
Evaluate which metrics can be produced reliably during simulated streaming.
Add more user interactions into our dashboard.
Additional dashboard for statistical analysis.

Questions?